

Electronic System Design Project Lab

Air Monitor

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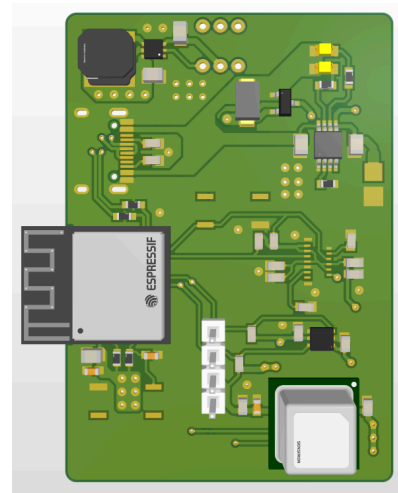
1 Features:

- Air Monitoring Board:
 - ▶ Input: 1000mAh Battery (Charging via USB), Ambient Air
 - ▶ Output: CO₂ (ppm), VOC Index, NO_x Index, PM
- Processing Unit: ESP32-C3-MINI
- Power System:
 - ▶ TP4056 Li-ion charging
 - ▶ RT8010A buck converter (VDD regulation)
- User Interface:
 - ▶ I₂C Display (real-time data)
 - ▶ Status LEDs
- Communication:
 - ▶ UART (debug/logging)
 - ▶ I₂C (sensors & display)

5 Measurement Specifications

- CO₂: $\pm(40 \text{ ppm} + 5\%)$
- VOC/NO_x: 10000 ppb
- PM: 1 $\mu\text{g}/\text{m}^3$
- Sampling Rate: 1 sample/10 sec

6 Draft PCB Layout



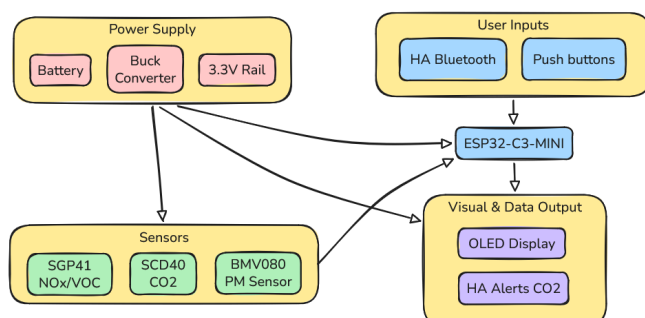
2 ICs Required:

1. BMV080: PM Sensor
2. SCD40: CO₂ Sensor
3. SGP41: NO_x/VOC Sensor
4. TC4056: Battery charger IC
5. ESP32-C3-MINI: MCU
6. RT8010/A: DC-DC Converter

3 Cost

- Components (Active+Passive): 6900 INR

4 Block Diagram



EE2504 - Project Idea

Local Climate Monitor

19-03-2026

Our Idea - Smart Air Quality Monitor

Problem :

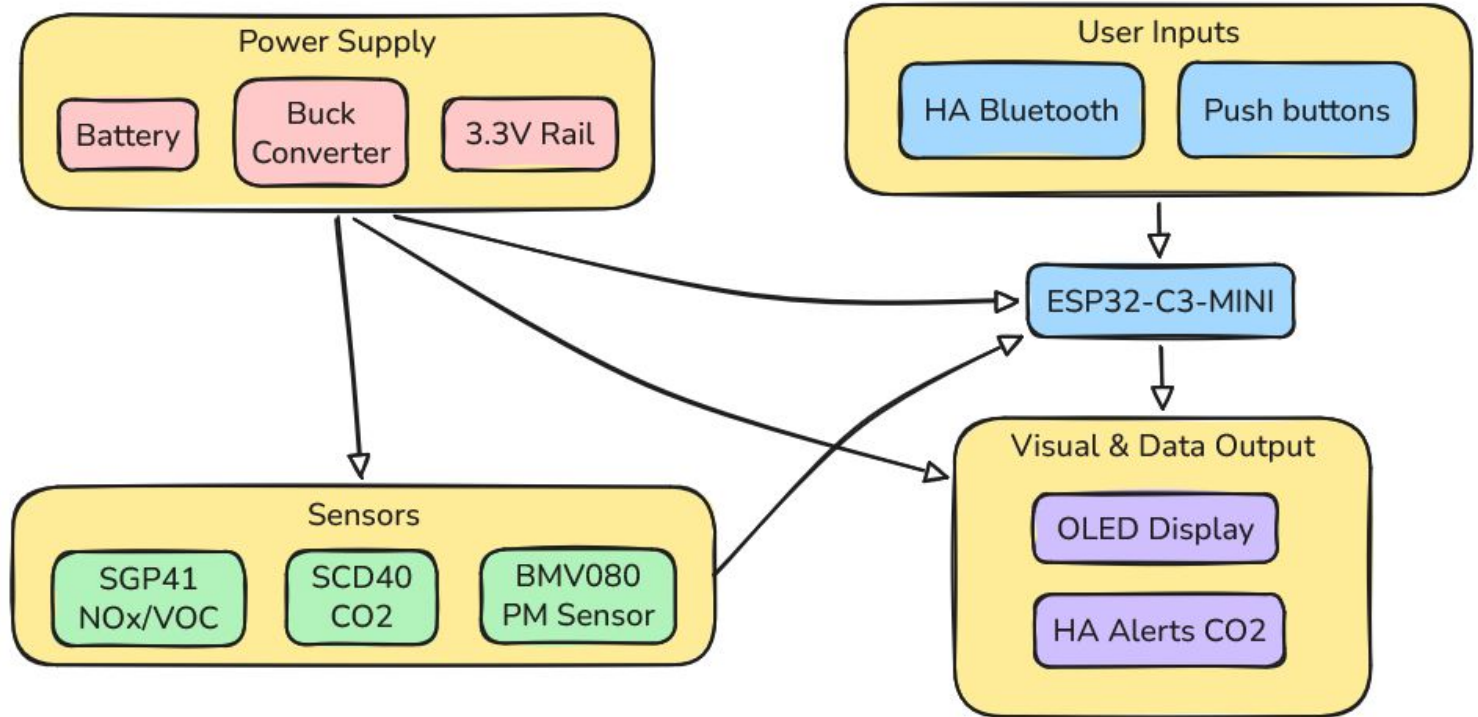
1. Most indoor workplaces are not monitored for potentially harmful environments.
2. If monitored also, not coupled with IoT for remote access / control of other IoT devices.

Our Idea - Smart Air Quality Monitor

Solution:

1. Constantly monitor local environmental parameters (air quality, CO2 levels, temperature, humidity)
2. Make remote access available (along with alerts in certain cases of abnormal changes in parameters)
3. Extension - use this data to control other IoT devices (example: controlling an air purifier fan based on AQI, controlling a fan based on temp/humidity)

Block diagram



Targets

- Realtime parameters monitoring using input of multiple sensors
- IoT integration with Home Assistant
 - Alerts (abnormal change in parameters)
 - Realtime Info

Sensors

- BMV080 sensor for measuring particulate matter
 - SCD40 for CO2 as well as measuring temperature and humidity
 - SGP41
 - Gas sensor - NOx , VOC etc..
-

Components and cost

SI No	Components' name	Components' Value	Manufacturer Part Number (MPN)	Package/Footprint	Quantity
1	BMV080 - PM Sensor	3500	BMV080	Flex PCB	1
2	SCD40 - CO2 Sensor	1500	SCD40-D-R2	SMD	1
3	SGP41	750	SGP41-D-R4	SMD	1
4	ESP32-C3FH4	300	ESP32-C3	Module - Mini	1
5	1000mAh LiPo Battery	370	WLY394058	Battery	1
6	TC4056 IC	12	TC4056	SMD - SOIC8	1
7	RT8010-33GQW Buck Converter	54	RT8010-33GQW	WDFN-6L	1
8	0.96" OLED Display	156	SSD1306	Display	1
9	Push Button - Latch - ON/OFF	5	KFC-5.8C	DIP-6	1
10	Push Button - Wake	5		SMD	3
11	10uF Capacitors	2.3	CL21B106KQNFNE	SMD - 0805	4
12	100nF Capacitors	2.3	CC0603KRX7R9BB104	SMD - 0603	4
13	1uF Capacitors	1.36	C0603C105K4RACTU	SMD - 0603	8
14	Connector for BMV080	87	503566-1302	SMD	1
15	USB-C Connector (16 pin - 2.0)	38	TYPE-C-31-M-12	SMD	1
16	4.7uF Capacitors	3	0805X475K500NT	SMD - 0805	1
17	2.2uH Inductors	23	NRS6028T2R2NMGJ	SMD	1
18	10k-ohm Resistors	1	RC0603FR-0710KL	SMD - 0603	2
19	0-ohm Resistors	1	RC0603FR-070RL/JR	SMD - 0603	2
20	5.1k-ohm Resistors	1.32	RC0603FR-075K1L/JR	SMD - 0603	2
21	4.7k-ohm Resistors	1	RC0603FR-074K7L	SMD - 0603	2
22	1.3k-ohm Resistors	0.15	AC0603FR-071K3L	SMD - 0603	3
23	LED-Red	1.18	Generic	SMD - 0805	2
24	LED-Green	2.62	Generic	SMD - 0805	2
25	Schottky Diode	1.35	SS14	SMA (DO-214AC)	1
26	PMOS IC	10	SI2305CDS-T1-GE3	SMD - SOT23	1

Thank you
